Practice 02

Note: Do not use arrays or nested loop in any task:

Q1: Nasir works in a factory. The company remains operative 24 hours. Nasir has flexible hours to finish his job. However, for the purpose of reporting the company has to record the number of hours each day he works. Further, the company has to record the total number of hours he worked in a week. You have to input all his working slots including start time and finish time (in 24 hours format). He may have one or more slots for each day. His slots may start in one day and end the next day. For example, he may start work at 22:00 (10 PM) and finish at 3:00 (3 AM) the next day.

The input is of five working days starting from Monday to Friday. The first input will be the number of total slots. For each slot, there will be two inputs that are start and finish time. There will be at least one input that will start from each day and there will be one input that will finish the next day. However, it is also possible that there will be more than one slot starting and finishing on the same day.

The slots will be non-over lapping. The minimum slot duration is one hour and maximum slot duration is 12 hours. Similarly, there will be a minimum one hour and maximum 12 hours gap between each slot. Each start will start and finish at the end of the hour only. You have to track each day from slots.

Input:

Number of slots: 10

Enter slot 1 start time: 7

Enter slot 1 finish time:: 10

Enter slot 2 start time: 18

Enter slot 2 finish time:: 2

Enter slot 3 start time: 9

Enter slot 3 finish time:: 14

Enter slot 4 starting time:: 3

...

Enter slot 10 start time: 16

Enter slot 10 finish time: 18

Explanation:

The first slot starts at 7am on the first day of the week and ends at 10 am on the same day. The second slot starts at 6pm (18 hours) on the first day because considering a gap of 32 hours until the second day is not possible. The second slot finishes at 2am on the second day due to two reasons. It cannot finish on the first day, in this case the finish time will be less than start time, which is not possible), and it cannot finish on the third day (in this case, there would be no starting and ending on the second day, which is not possible).

Again, the third slot starts at 9 am on the second day and finishes at 2 pm (14:00) on the same day. Therefore, for day 1, the working hours are 9, and 3 in the first slot, and 6 in the second slot from 6 pm to 12 am (midnight). For the second day, the working hours are 7, and 2 in the second slot, and 5 in the third slot.

Output:

Working Hours Day 1: 9

Working Hours Day 2: 7

Working Hours Day 3: ...

Working Hours Day 4: ...

Working Hours Day 5: ...

Total Working Hours: 42